

RESEARCH

Psychometric properties of the neighborhood characteristics scale for cardiovascular disease

Propiedades psicométricas de la escala características del vecindario para enfermedades cardiovasculares

Propriedades psicométricas da escala características da vizinhança para doenças cardiovasculares

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Abstract

Introduction: The community and neighborhood environment can act as a determining factor, both positive and negative, in the adoption of behaviors that affect health. The Neighborhood Characteristics Scale for Cardiovascular Disease measures environmental characteristics that are related to cardiovascular disease. **Objective:** To evaluate the psychometric properties of the Neighborhood Characteristics Scale for Cardiovascular Disease in University Students. **Methodology:** Quantitative, transversal and validation study. The sample consisted of 303 young university students from the nursing degree with ages of 18 to 24 years using simple random sampling. Young people who reported cardiovascular disease, were pregnant, and were taking psychotropic medications were not included. The neighborhood characteristics questionnaire for cardiovascular disease was used, and informed consent and data confidentiality were respected. Descriptive statistics, exploratory and confirmatory factor analysis were used; as well as Cronbach's α coefficient. **Results.** The majority were women with an average age of 19.6 years and with a progress of 18.1 % in the first semester. Bartlett's test of sphericity was statistically significant and the Kaizer-Meyer Olkin index with optimal sampling adequacy. Confirmatory factor analysis showed acceptable goodness-of-fit indices for a 15-item structure organized into three dimensions: Aesthetic quality, access to healthy food, and conducive environment for walking. The model demonstrated internal reliability of $\alpha = .873$. **Conclusions.** The proposed 15-item scale allows assessing the environment on cardiovascular diseases with a 3-factor factorial structure and good internal consistency.

Keywords: Heart disease risk factors; Validation study; Young adult (DeCS).

Resumen

Introducción: El entorno comunitario y vecinal puede actuar como un estímulo determinante, tanto positivo como negativo, en la adopción de comportamientos que afectan la salud. La escala de Características del Vecindario para Enfermedades Cardiovasculares permite medir cuáles son las características ambientales que están relacionadas con las Enfermedades cardiovasculares. **Objetivo:** Evaluar las propiedades psicométricas de la escala de características del vecindario para enfermedades cardiovasculares en jóvenes universitarios. **Metodología:** Estudio cuantitativo, transversal y de validación. La muestra estuvo conformada por 303 jóvenes universitarios de la licenciatura en enfermería con edades de 18 a 24 años utilizando un muestreo aleatorio simple. Se excluyeron a jóvenes que refirieron enfermedad cardiovascular, estar embarazadas y que tomaran medicamentos psicotrópicos. Se empleó el cuestionario de Características del Vecindario para Enfermedades Cardiovasculares, se respetó consentimiento informado y confidencialidad de datos. Se utilizó estadística descriptiva, análisis factorial exploratorio, confirmatorio; así como el coeficiente α de Cronbach. **Resultados:** La mayoría eran mujeres con promedio de 19.6 años, 18.1 % del primer semestre. La prueba de esfericidad de Bartlett fue estadísticamente significativa e índice de Kaizer -Meyer Olkin con adecuación muestral óptima. El análisis factorial confirmatorio mostró índices de bondad de ajuste aceptables para una estructura de 15 ítems organizados en tres dimensiones: calidad estética, acceso a alimentos saludables y ambiente para caminar. El modelo demostró una fiabilidad interna de $\alpha = .873$. **Conclusion:** La escala propuesta de 15 ítems permite evaluar el entorno sobre las enfermedades cardiovasculares con una estructura factorial de 3 factores y una buena consistencia interna.

Palabras clave: Factores de riesgo de enfermedad cardíaca; Estudio de validación; Adulto joven (DeCS).



Abstrato

Introdução: O ambiente comunitário e de vizinhança pode atuar como um fator determinante, tanto positivo quanto negativo, na adoção de comportamentos que afetam a saúde. A escala Características de Vizinhança para Doenças Cardiovasculares mede características ambientais relacionadas às doenças cardiovasculares. **Objetivo:** Avaliar as propriedades psicométricas da Escala de Características de Vizinhança para Doenças Cardiovasculares em Estudantes Universitários. **Metodologia:** Estudo qualitativo, transversal e de validação. A amostra foi composta por 303 jovens universitários de enfermagem, com idades entre 18 e 24 anos. Utilizou-se amostragem aleatória simples. Foram excluídos jovens que relataram doença cardiovascular, estavam grávidas e faziam uso de psicofármacos. Foi utilizado o questionário de características de vizinhança para doenças cardiovasculares, sendo respeitado o consentimento informado e a confidencialidade dos dados. Utilizou-se estatística descritiva e análise fatorial exploratória e confirmatória para a validade de constructo e o coeficiente α de Cronbach. **Resultados:** A maioria eram mulheres com média de 19.6 anos, 18.1 % do primeiro semestre. O teste de esfericidade de Bartlett foi estatisticamente significativo e o índice de Kaiser-Meyer Olkin com ótima adequação amostral. A análise fatorial confirmatória mostrou índices de ajuste aceitáveis para uma estrutura de 15 itens organizada em três dimensões: qualidade estética, acesso a alimentos saudáveis e ambiente propício para caminhadas. O modelo demonstrou confiabilidade interna de $\alpha = 0.873$. **Conclusões:** A escala proposta, com 15 itens, permite avaliar o ambiente em doenças cardiovasculares com uma estrutura fatorial de 3 fatores e boa consistência interna.

Palavras-chave: Fatores de risco de doenças cardíacas; estudo de validação; adulto jovem (DeCS).

Introduction

Cardiovascular diseases (CVD) are one of the leading causes of mortality and disability-adjusted life years. Epidemiological records for 2019 reported approximately 27 million deaths in developing economies ⁽¹⁾. In the Americas and the Caribbean, it ranks first among causes of death, with a prevalence of 164.67 deaths per 100,000 inhabitants ⁽²⁾. Recent studies have documented that the impact of CVD is increasing among young people, particularly those in college (between the ages of 18 and 30) ⁽³⁻⁷⁾. The onset of CVD is explained by a complex interaction between genetic predisposition and various environmental influences, which contribute to the progressive deterioration of cardiovascular tissue structures and functions ⁽⁸⁾.

In terms of environmental influences, it has been shown that the community and neighborhood environment can act as a decisive stimulus, both positive and negative, in the adoption of behaviors



that affect health ^(9,10). Several studies report that unfavorable neighborhoods have a negative impact on cardiovascular health through multiple interrelated factors. First, the high availability of processed foods and advertising of harmful products such as sugary drinks promote unhealthy eating patterns. The lack of pedestrian infrastructure and scarcity of green spaces limit opportunities for physical activity ⁽⁸⁻¹⁰⁾. Moreover, conditions such as insecurity in public spaces act as an additional barrier ⁽¹⁰⁻¹³⁾, discouraging mobility and the creative use of outdoor areas ⁽¹⁴⁻¹⁷⁾.

To estimate the community and neighborhood environment, various methods have been used, namely, censuses that reflect sociodemographic characteristics, rating the environment according to certain indicators ⁽⁸⁻¹⁰⁾ and the use of geographic information systems ⁽¹⁸⁻²²⁾, which characterize the availability and accessibility of resources in the neighborhood. Although these approaches provide relevant and complementary information, they have limitations in terms of validity and reliability, as they depend on self-reported data. In addition, the absence of updated censuses and variability in measurement criteria can reduce the accuracy and applicability of the results to specific contexts ⁽²³⁾.

To address these limitations, a Neighborhood Characteristics Scale for Cardiovascular Disease (ECVE) has been developed for Latino and African American populations ⁽²⁴⁾ that measures neighborhood characteristics in relation to CVD and evaluates four components: Aesthetic quality, conducive environment for walking, safety from crime, and availability of healthy foods.

The ECVE has undergone internal consistency and test-retest reliability testing, with acceptable reliabilities (aesthetic environment $\alpha = .89$, walking/exercise environment $\alpha = .78$, crime safety $\alpha = .82$, and access to healthy food $\alpha = .94$) ⁽²⁴⁾. It has been applied in the Mexican context considering only three of its dimensions in a sample of Mexican adults with Diabetes Mellitus II (DM2), where only the total reliability of the scale was reported ⁽²⁵⁾, leaving a critical breach in the university



population. This breach highlights two needs: The lack of instruments validated for the Mexican context, where cultural and sociodemographic factors modify the perception of the environment and the absence of specific tools for young people with different patterns of mobility and environmental perception.

Based on the above, it is important to examine how the scale works in young university populations in the Mexican context in order to make it valid and reliable for this group. This will help health professionals to better understand the impact of the environment in relation to CVD, to base interventions on evidence, and to strengthen primary prevention. Therefore, the objective of the study was to evaluate the psychometric properties of the Neighborhood Characteristics Scale for Cardiovascular Disease in young university students.

Methodology

Quantitative, cross-sectional study and validation ⁽²⁶⁾, a confirmatory factor analysis (CFA) and exploratory factor analysis (EFA) of the psychometric properties of the ECVE were performed. The sample consisted of 303 young university students from a public institution in Tamaulipas, Mexico, selected by simple random probability sampling using random number tables based on the official lists of the educational institution. The sample size was calculated according to the standard psychometric criterion of 5 to 10 observations per every item on the instrument ⁽²⁶⁾. University students enrolled in the first to eighth semester of the Bachelor of Nursing program, of both sexes, aged 18 to 24 years, and who submitted a signed informed consent form were included. Those who reported having any CVD, being pregnant, or taking psychotropic medications were not included. To collect information, a questionnaire was used that included variables such as age, gender, marital status, family history, and the ECVE, which consists of 27 items grouped into four factors: Aesthetic quality, conducive environment for walking, safety from crime, and availability of healthy food, with a Likert-type response format where 1 is strongly agree and 5 is strongly disagree



⁽²⁴⁾. The total score on the scale ranges from 27 points to 135 points ⁽²⁴⁾. The original version of the 27-item instrument has demonstrated good reliability with $\alpha = .88$ ⁽²⁴⁾. However, it is not valid for the population of young Mexican university students.

Data collection was carried out during school hours. Selected participants were invited to a space where they were provided instructions for filling the instrument.

Prior to participation, informed consent was obtained. To ensure anonymity, the questionnaires were placed in a sealed box. Those young people who did not wish to participate were thanked and given a leaflet with information on CVD prevention.

The study was approved by an ethics committee under registration number FAENUS-CEI-EXT-2024-02. In addition, it was regulated by the General Health Law on Health Research in Mexico, pursuant to Chapter I, Articles 13, 14, 16, 17, 21, 23, and Chapter V, which regulate dignified treatment, informed consent, data confidentiality, minimal risk research, dissemination of results, and reprisals for dropping out of the study ⁽²⁷⁾.

Data analysis was conducted using the Statistical Package for the Social Sciences (SPSS) and IBM-AMOS version 23. Descriptive statistics were used to describe the characteristics of the sample. Inference was performed using EFA, meeting the established criteria of Kaiser-Meyer-Olkin (KMO) $> .70$ and Bartlett's sphericity test of $X^2 p < .05$ (28,29). EFA was performed using the maximum likelihood extraction method with Varimax rotation. The criteria for factor membership were considered to be factor loadings > 0.30 and a minimum of three items for each factor. For CFA, the Root Mean Square Error of Approximation (RMSEA) $< .08$ was considered an adequate indicator of model fit (30). Standardized Root Mean Square Residual (SRMR) < 0.08 (31), Chi-Square Minimum/Degrees of Freedom (CMIN/DF) < 3.0 (32), Adjusted Goodness of Fit Index (AGFI) $> .90$, Comparative Fit Index (CFI) > 0.90 , and Comparative Fit Index (CFI) > 0.90 were used. Degrees of Freedom (CMIN/DF) < 3.0 (32), Adjusted Goodness of Fit Index (AGFI) $> .90$,



Comparative Fit Index (CFI) > 0.90, Goodness of Fit Index (GFI) > .90 (33,34), and Tucker Lewis Index (TLI) > .90 (35,36) were used. The reliability of the instrument and its dimensions was estimated using Cronbach's alpha coefficient.

Results

In terms of sociodemographic characteristics, the majority of the sample was made up by women (95 %) with an average age of 19.6 years (SD = 1.7), 95.0 % were single, and 18.1 % were in their first semester of a bachelor's degree in nursing. In terms of family history, DM2 predominated (41.7 %), and 16.7 % had a family member with more than two chronic diseases (DM2, high blood pressure, and obesity).

In the CFA, Bartlett's sphericity test was statistically significant, indicating sufficient correlations between the variables for CFA to be conducted. Likewise, the Kaiser-Meyer Olkin index showed a new optimal sample adequacy for factor analysis, supporting the validity of the factor model used, (Table 1).

Table 1. Criteria for exploratory factor analysis, 2024 (n= 303)

Kaiser-Meyer-Olkin sample adequacy measure		0.850
Bartlett sphericity test	Approximate chi-square	3409.45
	gl	351
	Significance	.001

Source: Self-development

Table 2 shows the factor loadings of the rotated AFE matrix, where three factors were identified: Aesthetic quality (8 items), healthy foods access (4 items), and conducive environment for walking (3 items), with an explained variance of 34.7 %. Likewise, the crime safety dimension was eliminated from the AFE because the items had factor loadings below < .30 and the explained variance was 5.9 %.



Table 2. Factor loadings of the rotated matrix, 2024 (n=303)

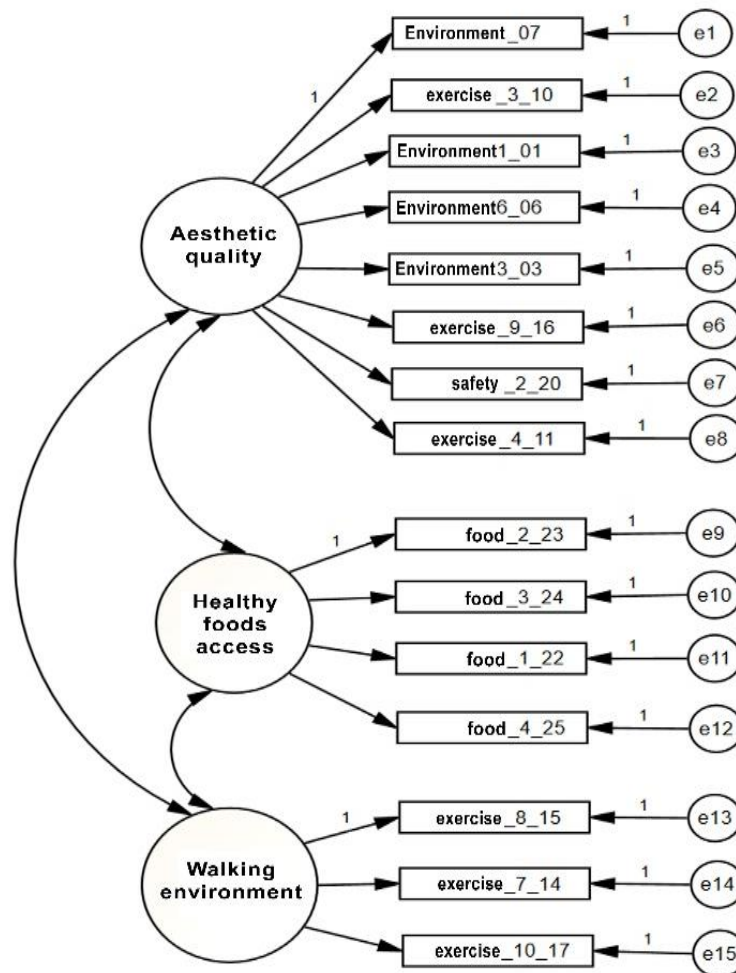
Item	Question	Factor load
Factor 1 Aesthetic quality		
Item 7	The houses and buildings are interesting.	.776
Item 10	It is pleasant to walk around my neighborhood.	.655
Item 1	My neighborhood is attractive.	.637
Item 6	The houses and buildings in my neighborhood are in good condition.	.594
Item 3	There are interesting things in my neighborhood.	.516
Item 16	The streets in my neighborhood are in good condition.	.473
Item 20	My neighborhood is crime-free.	.438
Item 11	There are enough trees in my neighborhood to provide shade.	.408
Factor 2 Healthy foods access		
Item 23	In my neighborhood, they sell a wide variety of fruits and vegetables.	.876
Item 24	The fruits and vegetables sold in my neighborhood are high quality.	.751
Item 22	In my neighborhood, it is easy to buy fresh fruits and vegetables.	.729
Item 25	In my neighborhood, it is easy to find low-fat products (milk, meats such as chicken, beef, and fish).	.559
Factor 4 Conducive environment for walking		
Item 15	The stores are close to my house.	.780
Item 14	It is easy to walk from my neighborhood to other places.	.439
Item 17	I often see other people walking around my neighborhood.	.321

Source: Self-development

The AFC evaluated the structure determined from the three correlated factors, confirming the presence of three latent factors: Aesthetic quality, healthy foods access, and conducive environment for walking, (Figure 1).



Figure 1. Neighborhood characteristics construct for cardiovascular disease, 2024 (n=303)



Source: Self-development

The ECVE construct comprised 15 items with good goodness-of-fit indexes and good reliability on the total scale of .873, (Table 3).

Table 3. Factor loadings, multiple correlation square value, and goodness of fit, 2024 (n=303)

Latent Factor	Item	Factor loading	R ²	α Cronbach	α Cronbach Total scale
Aesthetic quality	Item 7	1.00	.528	.826	.873
	Item 10	1.220	.572		
	Item 1	.975	.534		
	Item 6	.859	.348		
	Item 3	.839	.290		
	Item 16	.920	.364		
	Item 20	.717	.211		
	Item 11	.908	.277		
Healthy food access	Item 23	1.00	.872	.898	
	Item 24	.846	.741		
	Item 22	.897	.667		
	Item 25	.766	.511		
Conducive environment for walking	Item 15	1.00	.428	.687	
	Item 14	.884	.321		
	Item 17	.962	.504		
CMIN/DF		2.608			
GFI		.913			
AGFI		.880			
CFI		.928			
TLI		.913			
RMSEA		.073			
SRMR		.0535			
p		.001			

Source: Self-development. Note: CMIN/DF: Coefficient between the chi-square statistic and its degrees of freedom; CFI: Comparative fit index; GFI: Goodness-of-fit index; AGFI: Adjusted GFI; SRMR: Standardized residual mean square root; RSMEA: Root mean square error of approximation; p-value <.001.

Discussion

The objective of the study was to evaluate the psychometric properties of the ECVE in young university students. Based on the results, it was found that young university students shared similar characteristics, notably being single and having at least one first-degree relative diagnosed with a chronic disease, particularly DM2. These conclusions not only reflect a social and public health reality in Mexico, but also open the door to relevant discussions about the implications of these factors. The fact that most of the young people were single matches the demographic trends reported in Mexico by the National Institute of Statistics and Geography, which indicate that approximately more than three-quarters of people between the ages of 18 and 24 are single ⁽³⁷⁾.



This situation can be explained by the focus on academic and professional training during this stage of life, where priorities tend to be geared toward personal development and preparation for the labor market, postponing decisions about marriage or starting a family ⁽³⁸⁾.

Among the most significant findings of this study was the high prevalence of young people who reported having at least one immediate family member with DM2. This data is not surprising in Mexico, given that it ranks among the countries with the highest incidence of DM2 worldwide. The former data is consistent with data reported by the National Health and Nutrition Survey held in 2021 ⁽³⁹⁾, which reports that 14.6 % of Mexican adults have DM2, and it is estimated that just under a third of the population had a direct family member with this disease. The high prevalence of a family history of DM2 can be explained by a strong hereditary influence, but it also reflects the transmission of habits and lifestyles that favor the onset of this disease, such as unhealthy eating and a sedentary lifestyle. Likewise, the literature points out that young people with a family history of DM2 are not only at risk of developing the disease, but also that those who grow up in environments where DM2 is common may be more aware of the associated risks and adopt healthy lifestyles ^(40,41).

With regard to the ECVE, it did not replicate the original structure ⁽²⁴⁾, but rather a multidimensional type with three of the four original components and a total of 15 items. In turn, the validity of the construct was assessed based on the goodness-of-fit indicators obtained (CMIN/DF, GFI, CFI, RMSEA), which indicated an acceptable fit of the model and supported its adequacy. Although the SRMS value was below the cutoff point, this suggests that the differences between the observed and estimated covariance matrices are minimal ⁽⁴²⁾.

The first factor, called aesthetic quality, remained unchanged. However, there were items from the other subscales that contributed to this component, such as exercise and crime safety. Several causes could have led to this phenomenon. One possibility is that safety from crime is



interconnected with aesthetic quality because a well-maintained and visually appealing environment tends to provide a greater sense of security, which could explain the grouping of these items. The exercise items (items 10, 11, and 16) refer to green spaces, the state of repair of streets, and the community's perception of satisfaction with walking within their neighborhood, aspects that were closely linked to the aesthetic quality of the place ^(43,44). Another possible explanation for this grouping could be the extraction and rotation method used in the study, which may influence the final structure of the factors ⁽⁴²⁾.

The second component, healthy food access, retained most of the original items in accordance with the proposed theoretical structure. However, two items referring to the sale of reduced-fat products did not show significant loadings. This could be explained by cultural and perception factors, since young people in college, in particular, tend to perceive these types of products as expensive and less tasty ⁽⁴⁵⁻⁴⁷⁾.

The last component, referred to as environment, refers to the physical space designed or adapted to promote comfortable and safe walking. Three items from the original scale were grouped together in this dimension. This could be because the items are interrelated and consider aspects such as accessibility, connectivity between areas, and the presence of people in constant motion in dynamic environments, which are key characteristics of the conducive environment for walking ⁽⁴⁸⁻⁵⁰⁾.

The internal consistency of the model was acceptable ($\alpha = .873$), as were the components of aesthetic quality ($\alpha = .826$) and healthy food access ($\alpha = .898$). However, the third component, called conducive environment for walking, had low internal consistency. Two main factors could explain this result: First, the small number of items grouped under this factor, which limits the instrument's ability to capture the construct; second, the cross-cultural origin of the instrument, developed mainly in the United States for Latino and African American populations ⁽²⁴⁾ whose urban context



differs from Mexico's in key aspects such as pedestrian infrastructure, road safety, and urban design. These contextual differences may have affected the interpretation of the items by the population included in the study.

The results suggest that future research should adapt the conducive environment for walking component by incorporating new items that are culturally designed for the Mexican context. Likewise, variables such as pedestrian safety (lighting), pedestrian infrastructure (condition and accessibility of sidewalks, presence of obstacles), environmental factors (noise pollution), and road safety (signage and respect for pedestrians) could be considered to enrich this dimension.

This study offers a relevant tool for the prevention of CVD in young people in Mexico, although it is important to note some limitations. The research was conducted on a sample of young health science students in northern Mexico, which limits the results to other populations. Furthermore, the instrument was not culturally adapted. It is suggested that future studies include participants from different regions of the country and from different academic disciplines. It is recommended that the items in the original instrument be adapted semantically and idiomatically to ensure that they are clear, relevant, and culturally appropriate, in order to improve the internal consistency of the factors, especially those with low factor loadings.

Despite these limitations, the results obtained provide valuable preliminary data on the validation of the ECVE in young Mexican university students.

Conclusions

The final version of the ECVE in Spanish revealed adequate correlations, and the validation process confirms its applicability in the study population, meeting the validity criteria for each factor. The characteristics of the neighborhoods surrounding young university students are rarely evaluated because the prevalent age for CVD is over 30 years in older populations. Therefore, a scale for



evaluating the characteristics of the neighborhoods where young university students live plays a fundamental role in influencing health behaviors and exposure to cardiovascular risk factors.

This study contributes to behavioral sciences with a focus on healthy lifestyles, as it shows a psychometrically valid and abbreviated instrument that considers neighborhood characteristics as a health determinant. This determinant may influence the efficiency of measuring this phenomenon as a factor that relates to the behavior of young adult university students, considering health decisions such as healthy eating, engaging in physical activity that benefits cardiovascular health, and preventing chronic diseases.

The evidence identified in this study also contributes to science in terms of providing an empirical framework for future research in different contexts. With regard to practice, the results contribute to improving the certainty of the application of the neighborhood characteristics instrument for cardiovascular disease to guide the identification of environmental risk factors that affect cardiovascular risk based on community health needs, that is, specifically, characteristics related to the aesthetic quality of the environment or context, access and/or acquisition, as well as the consumption of healthy foods in conjunction with conducive environment for walking, which enable nursing professionals to characterize diagnoses that provide a profile of young adult university students in specific communities for timely intervention, using health policies aimed at the phenomenon studied in this research as a reference.

Conflict of interest

The authors declare that there are no conflicts of interest.

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Artificial intelligence

The authors declare that they have not used any type of artificial intelligence resource in any section of this manuscript.

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